

Minimal Change Disease

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INTRODUCTION

Minimal change disease (MCD) is the most common cause of nephrotic syndrome in children (approximately 80%) and accounts for 10% to 20% of cases in adults. MCD is so called because on kidney biopsy, glomeruli are normal or near normal on light microscopy, with podocyte foot process fusion or effacement found on electron microscopy. Compared to other causes of nephrotic syndrome, MCD has a higher rate of remission after steroid treatment, with better long-term kidney outcomes. However, MCD still causes significant morbidity, especially in older patients, due to complications such as acute kidney injury (AKI) and thrombosis. A high proportion of patients with MCD relapse, resulting in the need for multiple courses of steroids or alternative immunosuppressive therapy and significant associated side effects.¹

EPIDEMIOLOGY

The International Study of Kidney Disease in Children (ISKDC) study found that 363 of 471 (77%) children presenting with nephrotic syndrome had MCD as the underlying histologic diagnosis,² and MCD accounted for 94% of cases in patients younger than 6 years who responded to an 8-week course of oral steroids. In older children with nephrotic syndrome, the proportion with MCD decreases steadily, reaching about 50% of all cases between ages 8 and 16 years, with a rising proportion secondary to focal segmental glomerulosclerosis (FSGS) relative to MCD.

The incidence of MCD in children ranges from 1.2 to 7 new cases per 100,000 population. It has a higher incidence during winter and shows variability among different populations. The incidence of MCD is reported to be as low as 1 per million population in the United Kingdom and up to 27 per million in the United States. It is less common in Blacks, in whom nephrotic syndrome is much more likely to be caused by FSGS.³ In adults, MCD is generally rarer than in children, although incidence is less well defined.

ETIOLOGY

Most patients have idiopathic MCD. Secondary MCD is usually clinically indistinguishable from idiopathic MCD¹; causes include Hodgkin disease and nonsteroidal antiinflammatory drugs (NSAIDs) (see [Table 18.1](#) for full list).

Stimulation of the immune system (e.g., by infections or atopic reactions) has been linked with both onset and relapses of MCD. In children, most exacerbations occur with an upper respiratory infection,¹ and recent reports show an association with SARS-CoV-2. Atopic individuals have a higher risk of MCD, and atopic reactions, such as to pollen, bee stings, and food allergens, have been reported as triggers of MCD relapses. This clinical association led to the introduction of allergen-free diets for patients with MCD, and a gluten-free

diet, elemental diets, or skin desensitization may result in a decrease in proteinuria.⁴ A link between vaccinations and presentations and relapses of MCD has also been suggested.

Rare cases of familial MCD have been reported but likely represent an underlying mutation in genes encoding for slit diaphragm proteins instead of primary or idiopathic MCD.⁵ Certain human leukocyte antigen alleles and other genes have been suggested to lead to an increased risk of MCD.⁶

PATHOGENESIS

Traditionally, MCD has been considered an immune disorder in which circulating T cells release factors that increase the permeability of the glomerular filtration barrier to plasma proteins by altering podocyte shape. However, the pathogenesis remains poorly understood. [Fig. 18.1](#) summarizes proposed pathways involved in MCD.

Circulating Factors as Mediator of Disease

In 1974, Shalhoub postulated that MCD was a T cell–mediated disorder based on the lack of immune deposits in the glomerulus, the rapid response to steroids, the association with Hodgkin disease, and the observation that remission often occurred during resolution of measles infection.⁷ This hypothesis was supported by the observations that (1) supernatants from T cell hybridomas from a patient with MCD induced podocyte injury and proteinuria in rats,⁸ and (2) proteinuria remitted when kidneys from a patient with active MCD were transplanted into patients without nephrotic syndrome.⁹ Several candidate molecules have been considered as possible circulating factors.

Cytokines

Certain T cell cytokines have been proposed as circulating factors, including interleukins IL-4 and IL-13. Some animal studies have shown that IL-13 or IL-4 can lead to increased proteinuria and foot process effacement, although how this translates to human MCD is unclear.¹⁰ Levels of IL-4 and IL-13 are higher in children with nephrotic relapse compared with healthy controls or those in remission, although this is not true for all patients. IL-4 and IL-13 are both produced as part of a Th2 immune response, which underlies atopy. Cytokines from other T cells may also be implicated. IL-17 and IL-23 from proinflammatory Th17 cells are higher in MCD than controls, whereas IL-10 and TGF β from opposing T-regulatory (Treg) cells are higher in controls. An increased Th17:Treg ratio was found in MCD patients compared with a control group. This ratio normalized with therapy, leading to the hypothesis that MCD may be due to an imbalance of proinflammatory Th17 cells and immune-modulatory Tregs.¹¹ Other proinflammatory cytokines of the innate immune system may also have a role in MCD. IL-8 has also been shown to increase urine protein creatinine ratio after its infusion into rats.¹² Tumor necrosis factor- α (TNF- α) has also been shown to act directly on podocytes, where it causes nephrin

TABLE 18.1 Secondary Causes of Minimal Change Disease	
Causes	Examples
Drugs	NSAIDs, interferon- α , lithium (rare: usually causes chronic interstitial nephritis), tamoxifen, antimicrobials (ampicillin, rifampicin, cefixime), gold (rare: usually causes membranous nephropathy), mercury in skin-lightening products
Allergy	Pollen, house dust, insect stings, poison oak
Hematologic conditions	Hodgkin disease, non-Hodgkin lymphoma, leukemia, hemophagocytic syndrome, myelodysplastic syndrome
Solid organ tumors	Most commonly lung, colorectal, renal cell carcinoma, thymoma (rare: pancreatic, breast, bladder, ovarian)
Other	Immunizations

NSAIDs, Nonsteroidal antiinflammatory drugs.

downregulation, restructuring of the actin cytoskeleton, and altered focal adhesions.¹³ However, case studies show that TNF- α blockade with etanercept can also lead to MCD.

Hemopexin

Hemopexin is synthesized in the liver and is present in human plasma of patients with MCD. Mechanistically, hemopexin is thought to mediate podocyte injury in a nephrin-dependent manner, but the role of hemopexin in MCD remains unproven.

Microbial Products

A role for microbial products as circulating factors is attractive because of the association between upper respiratory infections and relapses. Two toll-like receptor (TLR) ligands, lipopolysaccharide (LPS), a component of the outer membrane of gram-negative bacteria, and polyinosinic-polycytidylic acid (polyIC), a viral-like particle, bind to TLR-4 and TLR-3 on podocytes, respectively, and cause proteinuria in animals.¹⁴ However, the relevance of these findings for human MCD remains unknown.

Glomerular Filtration Barrier in Healthy State and Minimal Change Disease During Relapse

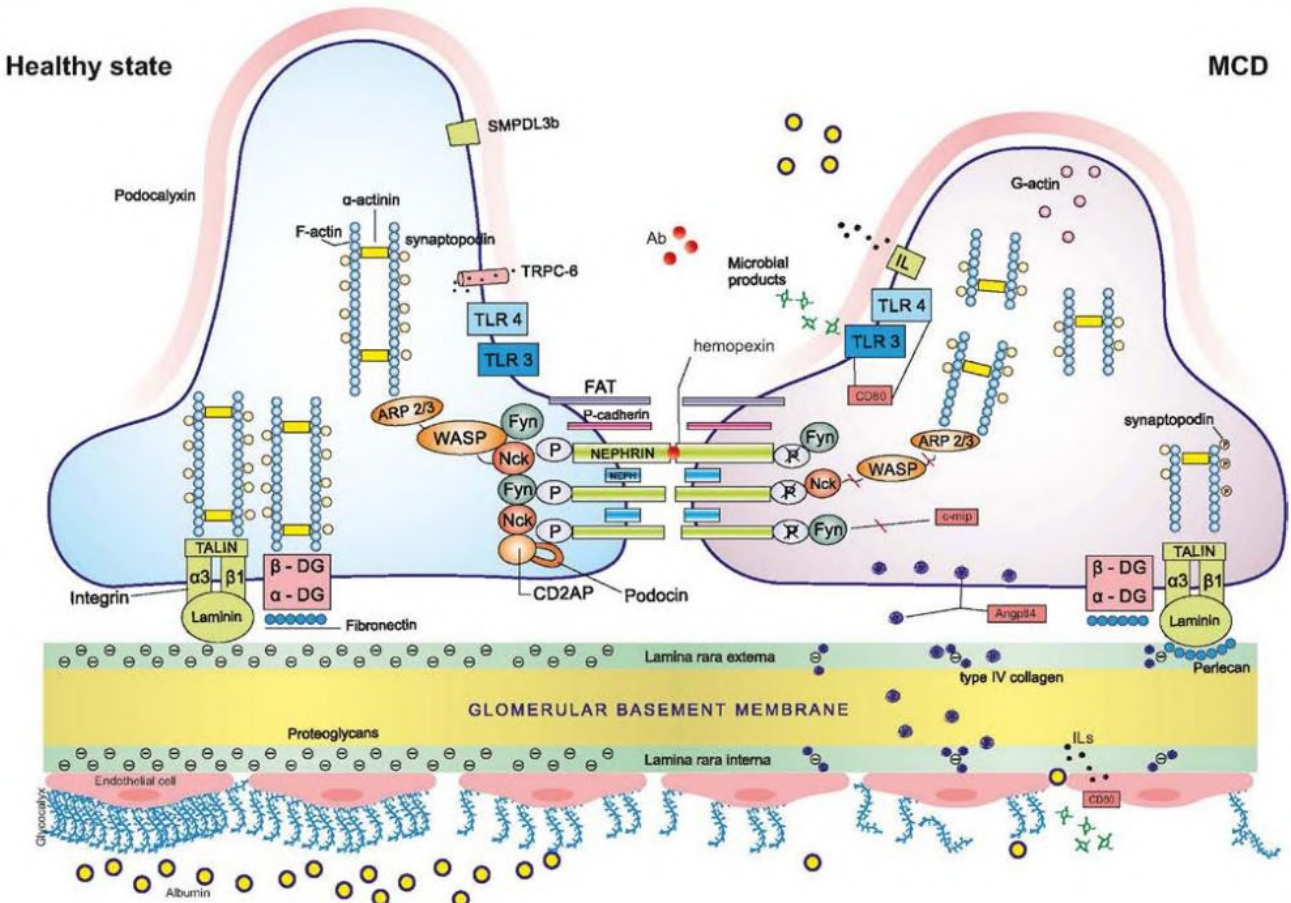


Fig. 18.1 Glomerular Filtration Barrier in Healthy State (*left*) and in Minimal Change Disease (MCD) During Relapse (*right*). In MCD, microbial products, circulating interleukins, hemopexin, and/or antinephrin antibodies (Ab) may target podocytes leading to foot process effacement and proteinuria. Circulating products may target and activate endothelial cells during relapse. Upregulation of CD80, c-mip, and angiotensin-like 4 (Angptl4) seem to play a role in experimental models of MCD-like injury. ARP 2/3, Actin-related protein 2/3 complex; CD80, cluster of differentiation 80; CD2AP, CD2-associated protein; c-mip, c-maf inducing protein; DG, dystroglycans; IL, interleukin receptor; SMPDL3b, sphingomyelin phosphodiesterase acid like 3B; TLR, toll-like receptor; TRPC6, Transient receptor potential cation channel subfamily C member 6; WASP, Wiskott-Aldrich syndrome protein. (Figure created using [biorender.com](#).)

Loss of Anionic Charges in the Glomerular Filtration Barrier

Heparan sulfate proteoglycans (HSPs) represent the major source of anionic sites in the glomerular basement membrane (GBM), and loss of HSP is observed in some MCD patients. Although HSP loss was thought to facilitate the passage of albumin through the GBM, subsequent experimental studies do not support this hypothesis driving proteinuria in MCD.¹⁵ An “electrokinetic model” has been proposed in that the endothelial glycocalyx represents a major site for the generation of a streaming potential that would repel albumin before passing through endothelial fenestration.¹⁶ Scattered reports showed increased circulating markers of endothelial injury in MCD, but their significance is unknown.

Podocyte Dysfunction as a Cause of Minimal Change Disease

In patients with MCD during relapse, podocytes display several molecular changes. An increase in podocyte focal adhesion kinase phosphorylation has been reported in MCD but not in FSGS.¹⁷ Likewise, changes in nephrin expression, phosphorylation, and localization have been reported by several groups.¹⁸ In experimental models, few molecules have been thought to mediate podocyte injury (Fig. 18.1).

CD80. In cultured podocytes, CD80 mediates actin reorganization and podocyte migration by targeting Neph 1 and $\beta 1$ integrins. In some patients with active MCD, CD80 can be upregulated in glomerular cells, primarily glomerular endothelial cells but also podocytes.¹⁹ In addition, most patients with MCD have higher urinary CD80 levels than patients with other glomerular disease,²⁰ but the relevance of this finding remains unknown. Anti-CD80 therapy has been associated with remission in two patients with MCD and high urinary CD80, whereas results in patients with FSGS have been contradictory.

Angptl4. This is a glycoprotein that has been proposed as a mediator of proteinuria in MCD. In a transgenic rat model characterized by glomerular overexpression of Angptl4 and podocin, rats had a marked loss of GBM heparan sulfate proteoglycans, podocyte foot process effacement, and albuminuria.²¹ However, data on Angptl4 in MCD are scarce and conflicting. A larger study did not confirm that MCD is associated with glomerular overexpression of Angptl4. In addition, urinary Angptl4 levels do not differ between different glomerular diseases.²²

C-mip. Experimental studies provide evidence for a pathogenic role of podocyte c-mip in proteinuria and podocyte injury. Mechanistically, c-mip impairs nephrin phosphorylation, which is a molecular feature of MCD. C-mip is upregulated in podocytes from patients with MCD but also in patients with other glomerular diseases,²³ so the exact role of c-mip in MCD is yet to be determined.

Role of B cells

The anti-CD20 monoclonal antibody rituximab has been effective for treating some patients with MCD, suggesting that B cells could be implicated in MCD. Recently, some patients with MCD were found to have antinephrin antibodies in circulation and in glomeruli.^{23b} Still, the role of B cells in MCD, either secreting antibodies or modulating T cell activation, remains poorly understood. It has been postulated that rituximab exerts a direct protective role in podocytes by binding to sphingomyelin phosphodiesterase acid-like 3 b (SMPDL3B),²⁴ but this remains controversial.¹⁰

In summary, the mechanisms of podocyte injury in MCD remain unknown. How findings from experimental studies may translate to human disease remain a matter of debate and clearly require further investigations. It also highlights the need to develop novel animal models of MCD. The observation that glomerular endothelial cells are activated during relapse warrants further investigation, as well as the role of B cells and B cell–T cell interactions in MCD.

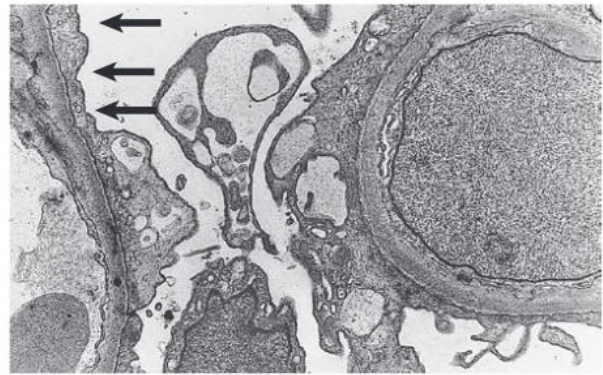


Fig. 18.2 Podocyte Foot Process Fusion and Effacement in Minimal Change Disease. The epithelial cells (arrows) are completely effaced along the glomerular basement membranes. (Electron micrograph; magnification $\times 6000$.) The normal appearance of epithelial cell foot processes is shown in Chapter 1, Figs. 1.6 and 1.7.

PATHOLOGY

MCD received its name because of the minimal, if any, glomerular abnormalities present by light microscopy. Mild changes such as a slight increase in mesangial matrix and hypercellularity may occasionally be observed. A small percentage ($<10\%$) of glomeruli may display global, but not segmental, glomerulosclerosis, which may reflect the natural senescence of glomeruli observed in young adults. In a few cases, minor focal tubular atrophy with segmental interstitial fibrosis may be present. Fat and hyaline droplets also may be found in the proximal tubule.

By immunofluorescence, immunoglobulins or complement deposits are rarely found. If present, deposits are limited to the mesangium. There is a small subset of patients who have IgM deposits, which have been considered by some clinicians to represent an entity distinct from MCD known as *IgM nephropathy* and characterized by a poorer clinical outcome. Although mesangial IgM had been considered due to “passive trapping” and of no pathogenic significance, subsequent studies suggested IgM may be directed against mesangial antigens and contribute to proteinuria.²⁵ Some case series have found mesangial C1q deposits, which may be associated with worse outcomes, leading to the concept of C1q nephropathy (see Chapter 29). However, it remains controversial whether the long-term prognosis is dictated by the presence of C1q deposits or, most likely, by the presence or lack of glomerulosclerosis on light microscopy.

Electron microscopy demonstrates diffuse podocyte foot process effacement or fusion (Fig. 18.2). Although fusion of foot processes is observed in other nephrotic conditions, in the absence of light microscopy glomerular changes, the finding is pathognomonic for MCD. The GBM is normal, and no electron-dense deposits are observed.

CLINICAL MANIFESTATIONS

Rapid onset of pitting edema, worse in the lower extremities, is often the first sign of MCD. Some patients present with bilateral periorbital edema, often misinterpreted as an allergic process. Edema may be mild or severe, presenting as anasarca with pleural effusions and ascites. Bowel edema may manifest as diarrhea. The mechanism(s) of edema formation in MCD is not fully understood but may be due to both a decrease in oncotic pressure and an increase in sodium and water reabsorption in the distal tubule (see Chapter 16).

Abdominal pain and nausea are common manifestations in MCD. Pain is usually dull due to massive ascites and bowel hypoperfusion. This pain resolves after albumin is administered. Severe pain may be due to peritonitis or acute pancreatitis.

MCD usually presents with a normal blood pressure. However, approximately 20% of adult patients are hypertensive on presentation. Some patients may develop transient renin-mediated hypertension during relapse from hypovolemia and renal hypoperfusion. Urinary changes may also be noted. Patients may describe “frothiness” of the urine. Macroscopic hematuria is concerning for renal vein thrombosis.

Less commonly, patients with MCD may develop “white nails,” sometimes in bands (Muehrcke lines) correlating with periods of clinical relapse (see Fig. 16.4). Adults may develop xanthomas and xanthelasma, especially on the eyelids (see Fig. 16.5).

LABORATORY TESTS

Nephrotic syndrome is the combination of nephrotic-range proteinuria, hypoalbuminemia, hyperlipidemia, and edema. In children, nephrotic-range proteinuria is defined as greater than 50 mg/kg/24 h, greater than 40 mg/h/m² (urinary specimens collected overnight), 200 mg protein/mmol urine creatinine, or a ratio of urinary protein to creatinine (uPCR) greater than 3 mg/mg.² In adolescents and adults, nephrotic-range proteinuria is defined as greater than 3.5 g/24 h or an uPCR greater than 3 mg/mg. In MCD, proteinuria is not only massive but also selectively albuminuria (see Chapter 16). Urine microscopy shows hyaline casts and fat bodies, as well as microscopic hematuria in 20% of patients.

Hypoalbuminemia is defined by a serum level of albumin below 2.5 g/dL, and it is the result of the increased glomerular filtration permeability to plasma proteins. The liver production of albumin is increased in MCD patients, and gastrointestinal losses are minimal. In MCD, massive edema usually develops with serum albumin levels below 2 g/dL.

Hyperlipidemia (elevated total cholesterol and low-density lipoprotein [LDL] cholesterol) is a universal feature of patients with MCD in relapse, and if massive proteinuria is persistent, triglycerides and very-low-density lipoprotein (VLDL) are also elevated. Hyperlipidemia results from increased hepatic synthesis of cholesterol and triglycerides, decreased activity of lipoprotein lipase, and altered expression of proprotein convertase subtilisin/kexin type 9 (PCSK9; see Chapter 16). In addition, urinary losses of lecithin cholesterol acyltransferase lead to a reduction of chylomicrons and VLDL clearance. The hyperlipidemia resolves after resolution of proteinuria.

Kidney function is commonly normal, but acute kidney injury is seen in up to 25% of adults. Oliguria can occur, and some patients may require dialysis, which can be prolonged in some cases. Risk factors for AKI include older age, history of hypertension, and more severe nephrotic syndrome. AKI is also seen in children hospitalized with MCD. AKI in MCD can be due to intravascular depletion, but without excessive diuretic therapy, intravascular volume is not reduced. Despite this, up to 60% of kidney biopsies in MCD patients with AKI show acute tubular injury, suggesting ischemia, the mechanism for which is not understood. One theory is that the nephrotic state in MCD causes local release of the potent vasoconstrictor endothelin-1.²⁶

Mild hyponatremia is common during relapse and may represent pseudohyponatremia due to hyperlipidemia. Total serum calcium may be low, as hypoalbuminemia leads to decreased protein-bound calcium. Ionized calcium is usually normal, and no treatment is indicated. Urinary losses of vitamin D-binding proteins can cause low vitamin D levels, and supplementation should be considered, especially if treatment with steroids is being considered.

Other Presentations (Complications)

Infection

Infections are a common complication of MCD, causing 19% of all hospital admissions for pediatric patients and a significant proportion of pediatric deaths. Mechanisms for increased susceptibility to

infection are diverse, including disruption of complement, especially in the alternative pathway where urinary losses reduce levels of factors B and I, persistent hypogammaglobulinemia, and immunosuppressive burden. Pneumonia, urinary tract infection, bacteremia, and peritonitis have all been reported. The predominant infective agents are *Streptococcus pneumoniae*, *Haemophilus* species, and *Escherichia coli*.²⁷

Thrombosis

The combination of hypercoagulability (decreased anti-thrombin III and proteins C and S and increased fibrinogen; coagulation factors V, VII, VIII, X, and XII; increased platelet aggregability) with hypovolemia (and resulting thrombocytosis) puts MCD patients at high risk for thrombosis. However, the risk is lower than in membranous glomerulonephritis (see Chapter 16). Thromboses are more common in adults than in children. Predisposing factors for development of thromboembolism include increased severity of the nephrotic syndrome at presentation, concurrent infection, male sex, and central venous catheterization. Presentations include pulmonary emboli, sagittal venous thrombosis, renal vein thrombosis, peripheral venous thrombosis, and less frequently, arterial thrombosis.²⁷ Children with nephrotic syndrome have a 2.8% chance of thromboembolism, whereas in adults that risk is 26.7%.²⁸

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

Kidney Biopsy

MCD is the most common type of idiopathic nephrotic syndrome in children, and therefore kidney biopsy is not routinely done in children younger than 10 years. However, if there are features that are not characteristic for MCD, such as hypertension, red cell casts, gross hematuria, kidney dysfunction, or lack of response to steroids (4- to 6-week course), a kidney biopsy should be considered at presentation.

In adults, a kidney biopsy is required to establish the diagnosis. Clinical and pathologic findings must be carefully interpreted. A patient with normal-appearing glomeruli who does not respond to immunosuppression could have a hereditary disease associated with a podocyte mutation (e.g., congenital nephrotic syndrome; see Chapter 20). Some patients with FSGS may only have a small percentage of glomeruli affected, which may not be evident on initial kidney biopsy sample due to sampling error. Indeed, there is ongoing debate about whether MCD and FSGS represent different diseases or whether they are the same disease at different stages.¹ MCD patients with steroid resistance often have findings consistent with FSGS in subsequent kidney biopsies.

NATURAL HISTORY OF MINIMAL CHANGE DISEASE

Children and adults with MCD classically present with nephrotic syndrome with normal kidney function, and spontaneous remission is uncommon in the absence of steroid treatment. Steroid treatment is usually effective in inducing remission, and time to remit with steroid therapy is longer with adults than children. In children, 50% of patients remit within 8 days of starting therapy, whereas adults have a median time to remission of over 2 months.

Up to 75% patients with both pediatric- and adult-onset MCD will have one or more relapses. Among children with MCD, 25% never relapse, 25% relapse infrequently, and 50% have numerous relapses. The last group are classified as frequent relapsers if they have at least four relapses per year and steroid dependent if relapse occurs during the steroid taper or soon after it is stopped. In adults, approximately 45% of patients have more than one relapse, a proportion of whom develop steroid dependence.²⁹ Relapses may occur at any point after

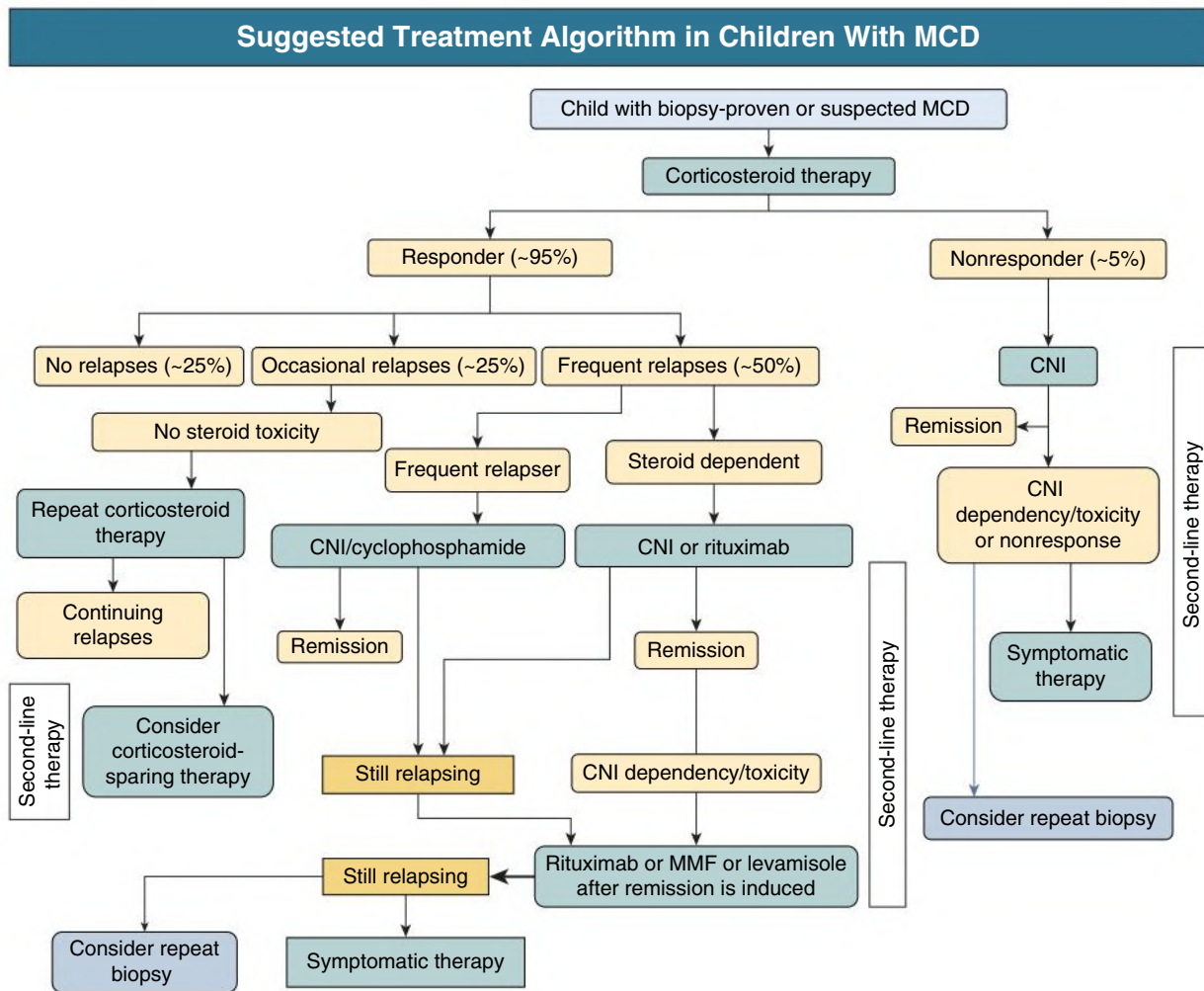


Fig. 18.3 Algorithm for Treatment of Childhood Minimal Change Disease. For definitions, see Table 18.2. The patient or parents should be involved in the decision after the potential side effects of the second-line treatment are considered in the rare patient who is a nonresponder to standard corticosteroid therapy and by definition is corticosteroid resistant. CNI, Calcineurin inhibitor; MMF, mycophenolate mofetil.

diagnosis, including at more than 15 years after initial presentation. However, patients in remission for more than 5 years may be less likely to relapse. There is no evidence that disease resolves during puberty, and it is not unusual to see children with MCD undergoing relapses during adulthood.

In contrast to FSGS, long-term kidney function in MCD is usually good. However, there is significant morbidity, mortality, and reduced quality of life, from both relapse and treatment, especially in older adults.

TREATMENT

General Considerations

Management of nephrotic syndrome is discussed in Chapter 16, including a low-sodium diet to control edema. Diuretics are infrequently used in children because of potential further volume depletion but are often used to control extracellular fluid volume in adults, in whom hypovolemia before treatment is less common. For patients who are diuretic resistant, intravenous albumin is frequently given, although there is scarce evidence for benefit.³⁰ Prophylactic anticoagulation should be considered for patients with pronounced hypoalbuminemia (i.e., serum albumin <2 g/dL), particularly in adults whose nephrotic

syndrome is likely to be more prolonged. One reported regimen uses daily low-molecular-weight heparin or warfarin if serum albumin is below 2 g/dL, switching to aspirin once serum albumin rises to over 2 g/dL and the thrombosis risk is reduced.³¹ There is little data on the use of newer direct oral anticoagulants, such as apixaban and rivaroxaban, in nephrotic syndrome, but these agents are highly protein bound, leading to increased urinary losses that may reduce efficacy.³²

Management of AKI in MCD is supportive. Patients with persistent AKI should be considered for repeat kidney biopsy.

Treatments for hyperlipidemia are more commonly given in adults than children due to the increased time to remission. Statins are the mainstay of cholesterol-lowering therapy, although these have little effect on triglyceride or lipoprotein levels.

Treatment in Children

A summary of suggested treatment for children with MCD is shown in Fig. 18.3.

Initial Treatment

Steroids are the first-line therapy to induce remission in children with idiopathic nephrotic syndrome and biopsy-proven MCD, because 95% of MCD patients respond to this therapy. The ISKDC recommended

TABLE 18.2 Terms Used in Nephrotic Syndrome in Adults and Children

Classification	Definition
Nephrotic syndrome	Nephrotic-range proteinuria (uPCR ≥ 2 mg/mg or ≥ 200 mg/mmol or 3+ protein on urine dipstick for children, or ≥ 3.5 g/day proteinuria for adults) and either hypoalbuminemia ≤ 3 g/dL (≤ 30 g/L) or edema when albumin level is not available
Complete remission	uPCR < 0.2 mg/mg or < 20 mg/mmol or negative or trace dipstick on three or more consecutive occasions (children) Reduction of proteinuria to ≤ 0.20 g/day and serum albumin > 3.5 g/dL (adults)
Partial remission	uPCR > 0.2 but < 2 mg/mg or > 20 and < 200 mg/mmol, and, if available, serum albumin ≥ 3 (children) Reduction of proteinuria to between 0.21 g/day and 3.4 g/day $\pm$ decrease in proteinuria of $\geq 50\%$ from baseline (adults)
Relapse	Recurrence of nephrotic-range proteinuria. In children, relapse is commonly assessed by urine dipstick, defined as dipstick $\geq 3+$ for 3 consecutive days or $\geq 1+$ for 7 days. In adults, relapse is defined as proteinuria ≥ 3.5 g/day occurring after complete remission obtained for > 1 month.
SSNS	Complete remission after 4 weeks of prednisone or prednisolone at standard dose (children)
IRNS	One relapse per 6 months or less than four relapses per 12 months
FRNS	Two or more relapses per 6 months, or four or more relapses per 12 months
SDNS	Relapses during therapy with prednisone or prednisolone (either at full dose or during tapering) or within 15 days of prednisone or prednisolone discontinuation
SRNS	Lack of complete remission at 4 weeks of therapy with daily prednisone or prednisolone at standard dose (children) Persistence of proteinuria despite prednisone therapy, 1 mg/kg/day for 16 weeks (adults)
Initial responder	Attainment of complete remission within initial 4 weeks of corticosteroid therapy
Late responder	Complete remission at 6 weeks (children)
CNI-responsive SRNS	Partial remission after 6 months of treatment and/or complete remission after 12 months of treatment with a calcineurin inhibitor at adequate doses and/or levels
CNI-resistant SRNS	Absence of partial remission after 6 months of treatment with a calcineurin inhibitor at adequate doses and/or levels
Multidrug-resistant SRNS	Absence of complete remission after 12 months of treatment with two mechanistically distinct steroid-sparing agents at standard doses
Secondary SRNS	A SSNS patient at disease onset who at a subsequent relapse fails to achieve remission after 4 weeks of therapy with daily prednisone or prednisolone at standard dose

CNI, Calcineurin inhibitor; FRNS, frequent-relapsing nephrotic syndrome; IRNS, infrequent-relapsing nephrotic syndrome; SDNS, steroid-dependent nephrotic syndrome; SRNS, steroid-resistant nephrotic syndrome; SSNS, steroid-sensitive nephrotic syndrome; uPCR, urinary protein to creatinine ratio.

an induction phase consisting of prednisone 60 mg/day/m² (equivalent to 2 mg/kg, with a maximum of 60 mg/day for children and 80 mg/day for adolescents) for 4 weeks.² Most nephrologists administer 2 mg/kg of prednisone as a single morning daily dose based on evidence that it provides equal efficacy than three times a day, and it is associated with less adrenal insufficiency and better adherence.

Although the German Arbeitsgemeinschaft für Pädiatrische Nephrologie reported a greater rate of sustained remission in patients treated with extended versus short (6 vs. 4 weeks) therapy with steroids,³³ this was not confirmed by the Southwest Pediatric Nephrology Study Group.³⁴ Given these results, the 2021 Kidney Disease: Improving Global Outcomes (KDIGO) guidelines recommend treatment of the initial episode of nephrotic syndrome in children with daily oral prednisone or prednisolone at 60 mg/m²/day or 2 mg/kg/day (maximum 60 mg/day) as induction therapy for 4 to 6 weeks.³⁵

Most children enter remission during the induction phase, but treatment is continued to the 4-week or 6-week time point, and then prednisone is slowly tapered to reduce the risk of side effects. During the tapering phase, prednisone is administered on alternate days, but there is no agreement on the duration or dose.

The 2021 KDIGO guidelines recommended oral prednisone at 40 mg/m² or 1.5 mg/kg on alternate days and continued for another 4 to 6 weeks.³⁵ A Cochrane systematic review in 2015 concluded that a 6-month prednisone course does not reduce the risk of relapse compared with a 2- or 3-month course in children aged 1 to 17 years at presentation. More recently, the PREDNOS study found no difference in time or number of relapses or relapse pattern among

children who received standard (4-week course with prednisone 40 mg/m² on alternate days) or long taper regimen (12-week therapy starting at 60 mg/m² on alternate days and tapering by 10 mg/m² every 2 weeks).³⁶ Thus, the 2020 Cochrane update concluded that there is no benefit of prolonging prednisone therapy beyond 2 to 3 months in the first episode of steroid sensitive nephrotic syndrome (SSNS).³⁷

Treatment of Relapses

There is no consensus on the prednisone regimen for induction and tapering phases. The ISKDC recommends prednisone at 60 mg/m²/day until response (maximum of 4 weeks) followed by prednisone 40 mg/m²/day for 3 consecutive days in a week for a total of 4 weeks.² Recently, the PROPINE trial found no differences in outcomes among patients who, after remission, received a short (18 doses of prednisone 40 mg/m² on alternate days) versus extended taper (prednisone at 40 mg/m² dose on alternate days tapered over 72 days by steps of six doses).³⁸ Other groups have studied the efficacy of different doses of steroids. In a prospective study, Borovitz and colleagues showed that children receiving prednisone at 1 mg/kg/day and followed by stepwise taper had similar rate of remission and relapse rate than those treated with a standard regimen (prednisone 2 mg/kg/day until remission followed by slow taper over 10–12 weeks).³⁹

The ISKDC and KDIGO provided arbitrary definitions regarding the relapse pattern, grouping patients as steroid dependent (SDNS) and frequent relapsers (FRNS) (see Table 18.2).^{2,35} These are widely used, but the distinction between them should be made carefully to

select appropriate therapy based on the available studies because definitions of steroid dependence can vary among different centers.

Steroid-Sparing Therapies

Whereas initial treatment of relapsing patients usually involves repeated courses of steroids, long-term exposure to steroids can cause significant toxicity, including behavioral changes, growth impairment, cataracts, hypertension, and diabetes among others. Therefore, alternative therapies are often required in children with MCD who are steroid dependent and/or relapse frequently. Patients should be ideally in remission prior to the initiation of any steroid-sparing agent. This is particularly important when using cyclophosphamide to minimize the risk of hemorrhagic cystitis. There is no consensus on the best second-line therapy for children with MCD.

Cyclophosphamide

This alkylating agent has been used in MCD for decades and used to be the most common first-line steroid-sparing agent. Cyclophosphamide can induce sustained remission, particularly in patients with FRNS, whereas its efficacy is lower in those with SDNS.³⁵ The optimal length of cyclophosphamide therapy is debatable and varies from 8 to 12 weeks. The most common side effect is bone marrow suppression, so weekly cell blood count is recommended, with adjustment of dosage or cessation of therapy if necessary. Second courses of cyclophosphamide should not be given (maximum cumulative dose of 168 mg/kg) to minimize gonadal toxicity. Some centers may favor other newer therapies due to a theoretical safer profile.

Calcineurin Inhibitors: Cyclosporine and Tacrolimus

Treatment with cyclosporine results in a similar rate of remission compared with cyclophosphamide and chlorambucil. However, its long-term efficacy is hampered by frequent relapse shortly after calcineurin inhibitor (CNI) withdrawal. Tacrolimus appears to have efficacy similar to that of cyclosporine.⁴⁰ The recommended dose for cyclosporin and tacrolimus are 4 to 5 mg/kg/day (in two divided doses) and 0.1 mg/kg/day (in two divided doses) with target 12-hour trough levels of 60 to 150 ng/mL and 5 to 10 ng/mL, respectively, aiming for the lowest levels to maintain remission and avoid toxicity. These recommendations are largely based on the experience in kidney transplantation, but have not been formally validated in MCD. The length of therapy varies from 12 to 24 months.³⁵ Mild to moderate cyclosporine-associated nephrotoxicity has been reported in up to one-third of MCD patients treated with cyclosporine for more than 3 years.

Rituximab

Rituximab has been used in MCD for over a decade. The 2021 KDIGO Guidelines and the 2020 Cochrane update recommend rituximab as a steroid-sparing agent.^{35,37} Iijima and colleagues found a significantly longer relapse-free period and fewer relapses in patients who received rituximab compared with those receiving a placebo.⁴¹ A recent randomized controlled trial (RCT) showed that rituximab was associated with a higher 12-month relapse-free survival rate than tacrolimus (90% vs. 63.3%, respectively) in children with SDNS, suggesting that rituximab could potentially be used as a first-line steroid-sparing agent in these children.⁴² Overall, rituximab is well tolerated. Infusion reactions are the most common adverse event. Serious complications include rituximab-associated lung injury, colitis, multifocal leukoencephalopathy, persistent hypogammaglobulinemia, and infections.

Mycophenolate Mofetil

Uncontrolled studies suggested beneficial effects of mycophenolate mofetil (MMF) therapy, but this has not been confirmed in RCT.

Dorresteijn and colleagues found that children receiving MMF had a higher risk for relapse than those on cyclosporine, though the difference did not reach statistical significance.⁴³ In a larger crossover trial using MMF or cyclosporine, patients on MMF suffered more relapses than those on cyclosporine, though patients with higher plasma levels of MMF (area under the curve >50 µg/h/mL) had a similar course to those treated with cyclosporine.⁴⁴ MMF is a relatively safe drug that can be associated with mild and dose-dependent side effects (gastrointestinal, bone marrow suppression) but not nephrotoxicity. Therefore, MMF seems a reasonable alternative as a steroid-sparing drug, especially if plasma levels are closely monitored. However, MMF is teratogenic, so caution is required when used in young females.

Other Medications

Levamisole. One RCT showed sustained remission in 26% of patients treated with levamisole compared with 6% in the placebo group.⁴⁵ Levamisole was more effective in patients from India than Europe. It is unclear whether this is because SDNS was more common in Europe or because helminthic infections could trigger MCD. Still, the use of levamisole is limited as it is not available in some countries and because it is associated with agranulocytosis and antineutrophil cytoplasmic antibody vasculitis.

Adrenocorticotrophic hormone. The ATLANTIS trial found a similar rate of relapse in children with FRNS and SDNS who received adrenocorticotrophic hormone (ACTH) therapy versus no relapse-preventing treatment,⁴⁶ so there is no evidence to support the use of ACTH in children with MCD.

Treatment in Adults

Initial Treatment

A summary of treatment recommendations given here is shown in Fig. 18.4. Although a small proportion of patients with MCD will spontaneously remit, withholding immunosuppression has been shown to be associated with increased risk of acute kidney injury and steroid resistance.⁴⁷

Steroids

Steroids are widely used as the initial therapy for MCD. Studies comparing different steroid regimens in adults with MCD are limited and regimens are extrapolated from successful approaches in children with steroid-sensitive nephrotic syndrome, although often with slightly lower doses of oral prednisolone (1 mg/kg/day, up to maximum 80 mg/day).⁴⁸ There is no good evidence that alternate-day steroids (prednisolone 2 mg/kg, up to maximum 120 mg) offer any clinical advantages over daily dosing; however, induction with methylprednisolone pulses may lead to more rapid responses and fewer relapses.⁴⁹

Response is often delayed in comparison with that in children, with a median time to remission of 10 weeks, and 25% fail to remit after 3 to 4 months.⁵⁰ Although unclear, the reason may be that adults are often given a relatively lower dose of steroids or that a greater proportion of adults have FSGS, missed on the original biopsy, which is more likely to be steroid resistant (SR). More recent work has shown a correlation between reduced nephron mass and increased time to remission with steroids.⁵¹ The KDIGO guideline recommends that the duration of steroid treatment should be up to 6 months, but this is based on very low-grade evidence.³⁵ The initial high dose of steroids, if tolerated, should be maintained for a minimum of 2 weeks after complete remission is achieved and for a maximum of 16 weeks if complete remission is not achieved. After this, steroids should be tapered over 24 weeks, although the rate of dose tapering and total length of treatment of the initial episode may need to be reduced if steroid toxicity is significant. If the patient has not responded after 12 to 16 weeks, genetic screening for FSGS or a repeat biopsy should be considered.

Suggested Treatment Algorithm in Adults With MCD

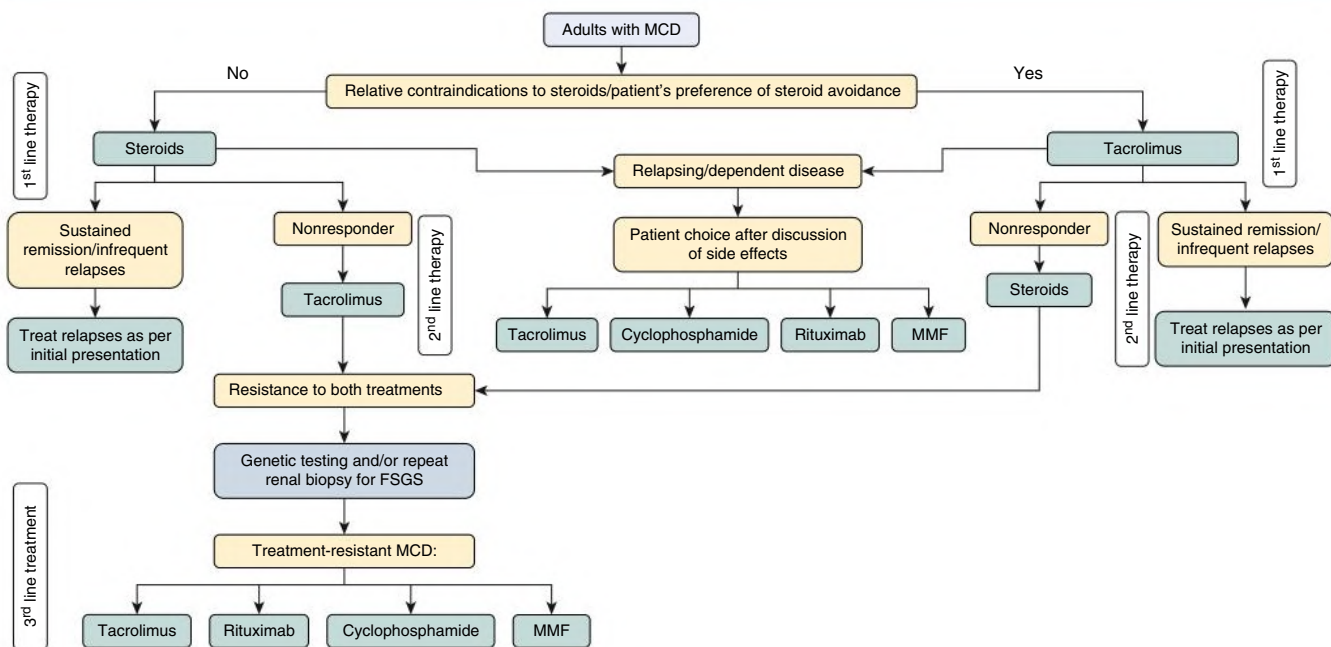


Fig. 18.4 Algorithm for Treatment of Adult Minimal Change Disease (MCD). FSGS, Focal segmental glomerulosclerosis; MMF, mycophenolate mofetil.

Infrequent relapses should be treated in the same way as the initial presentation, but there is even less evidence that a prolonged course of steroids is beneficial in reducing the frequency of subsequent relapse.

Although steroids are a highly effective treatment for MCD, prolonged courses of steroids often cause significant side effects. Up to 90% of adults taking steroids for more than 60 days report a side effect of steroid therapy; weight gain is the most commonly reported alongside skin changes, muscle weakness, sleep disturbance, cataracts, hyperglycemia, psychiatric disturbance, and bone demineralization leading to pathologic fractures. In other areas of nephrology such as kidney transplantation, steroid avoidance and early withdrawal protocols have led to improved outcomes, and similar strategies have been studied in MCD.

Calcineurin Inhibitors

A large RCT compared intravenous methylprednisolone followed by oral tacrolimus with oral prednisolone as initial therapy for MCD. This study found a similar rate of remission and relapse between both groups, but the oral prednisolone arm had a worse side effect profile.⁵² A more recent RCT compared low-dose prednisolone and tacrolimus with higher-dose prednisolone monotherapy. They found comparable rates of complete and partial remission but a significantly lower rate of relapse in the combination therapy group compared with steroids alone. This study showed a comparable rate of adverse events across both groups.⁵³

Use of tacrolimus monotherapy with complete avoidance of steroids has also been shown to be effective. The MINTAC trial involved 55 adults who were given either tacrolimus or prednisolone as initial therapy. There was no significant difference in complete or partial remission rates after 8, 16, and 26 weeks of treatment. There was no significant difference in relapse rates and no significant difference in the rate of adverse events.⁵⁴ This may be due to the relatively conservative steroid doses used in this trial. A second RCT also comparing tacrolimus with prednisolone as first-line MCD treatment also found similar remission and relapse rates, but adverse events were significantly more

common in the steroid-treated group.⁵⁵ Tacrolimus may be particularly useful for patients with contraindications to steroids, such as diabetes or psychosis.

Mycophenolate Mofetil

MMF has also been tried as a steroid sparing agent for initial therapy. Rémy and colleagues compared MMF plus low-dose prednisolone (0.5 mg/kg/day) with standard-dose prednisolone (1 mg/kg/day) alone in 116 adult patients and found no significant difference in rates of remission or relapse and no benefit in adverse events between the two treatment arms.⁵⁶

Cyclophosphamide

Cyclophosphamide can also be used as initial therapy in MCD instead of steroids,⁴⁸ although there are no RCTs to show efficacy. This may be suitable in older adults, where there is less concern regarding impacts on fertility or increased lifetime malignancy risk.

Rituximab

Other alternatives for first-line therapy include rituximab, although there is still a paucity of evidence relating to this. Fenoglio and colleagues gave rituximab as a first-line therapy to treatment-naïve adult patients with MCD who had contraindications to steroid therapy; five out of six patients achieved complete remission and did not relapse until the end of the follow-up period. The remaining patient achieved partial remission.⁵⁷ Patients may require enhanced dosing if rituximab is given when nephrotic due to protein binding of the drug.

Treatment of Relapses and Steroid-Sparing Therapies

As in children, some adults have transient nonnephrotic relapses, so it is important to establish that a full relapse has developed with persistent nephrotic range proteinuria before recommencing immunosuppression. Several different immunosuppressant therapies are effective for relapsing MCD; hence the choice of treatment should be tailored to individual patient's needs.

The CNIs cyclosporine and tacrolimus have similar efficacy, inducing remission in up to 75% of patients, but as in children, relapse often follows dose reduction or withdrawal. Due to its side effect profile, tacrolimus may be preferred to cyclosporine.⁵⁸ Initial dosing is 0.05 to 0.1 mg/kg, with target tacrolimus levels of 4 to 8 ng/mL, but higher levels may be required to maintain remission. Tacrolimus is suitable for young adults wishing to preserve fertility and is safe in both pregnancy and breastfeeding.⁵⁹ Recommended dosing for cyclosporine is 4 to 6 mg/kg/day, aiming for trough whole-blood levels of 150 to 200 ng/mL. KDIGO guidelines recommend CNI therapy should continue for 1 to 2 years,⁴⁸ but some patients require longer or repeated courses of therapy.

Rituximab can reduce the frequency of relapses and need for concomitant immunosuppression,⁶⁰ although definitive RCTs in adults are awaited. Median time to relapse with rituximab has been reported as 18 months.⁶¹ Longer-term maintenance dosing of rituximab can be guided by B cell monitoring, but relapse can recur quickly after repletion. For these patients, regular six-monthly dosing may be preferable and has been shown to be effective.⁶² Rituximab is generally well tolerated in adults with MCD, but patients need to be monitored for hypogammaglobulinemia. For patients who struggle with adherence to a daily oral medication, rituximab may offer some advantage due to its administration by intermittent infusion. Optimal doses and schedule remain to be determined.²⁴

A 12-week course of oral cyclophosphamide 2 to 2.5 mg/kg/day induces a longer-term remission more often in adults than children (75% and 66% at 2 and 5 years, respectively).⁶³ Although there are no satisfactory studies comparing 8-week and 12-week courses in adults, the 12-week course is logical based on the pediatric experience. If this treatment is selected, the opportunity for banking sperm or retrieval of ova should be considered before treatment in appropriate patients. Intravenous cyclophosphamide has been recommended to limit the cumulative dose: intravenous cyclophosphamide 0.5 g/m² per month adjusted upward to 1 g/m² based on the leukocyte count after 2 weeks with a target nadir of 3000 cells/mm³.

Several uncontrolled reports suggest that MMF is effective in inducing remission in steroid- and cyclosporine-dependent patients²⁹ and is generally well tolerated. However, it may not be suitable in young women due to teratogenicity.

Steroid Resistant Disease

For the 10% to 30% adults who do not remit with primary therapy, treatment of MCD may be challenging. For patients not remitting on steroids, tacrolimus can be used as second-line therapy, and vice versa for patients failing to remit on tacrolimus. Patients should be offered genetic testing or repeat biopsy to look for FSGS. For patients with truly SR MCD, case series have shown that CNIs, cyclophosphamide,⁶⁴ MMF,⁶⁵ and rituximab⁶⁶ can all induce remission.

MINIMAL CHANGE DISEASE WITH NONNEPHROTIC PROTEINURIA

It is possible to have MCD with subnephrotic proteinuria. These patients may be treated with antiproteinuric medications such as angiotensin converting enzyme inhibitors or angiotensin receptor blockers. If there is any uncertainty regarding diagnosis, a repeat biopsy should be offered.

Treatment of Secondary Minimal Change Disease

MCD secondary to NSAIDs requires discontinuation of the offending medication. Many patients are treated with a course of steroids for MCD (higher dose) or for acute interstitial nephritis (see [Chapter 64](#)), but evidence of benefit is uncertain. Secondary MCD from Hodgkin disease usually responds to treatment of the lymphoma. Some patients will also receive a drug regimen for MCD as adjunctive therapy in addition to the chemotherapy directed at the tumor in particular if MCD remission does not occur quickly with adequate chemotherapy.

SELF-ASSESSMENT QUESTIONS

- Which of the following statements regarding the presentation of MCD is *correct*?
 - Less than 1% of adults present with AKI.
 - Microscopic hematuria excludes a diagnosis of MCD.
 - Edema is rarely the first presenting feature.
 - Children may present with abdominal pain, ascites, and diarrhea.
 - Urinary frothiness indicates a patient has achieved remission.
- Which of the following statements regarding the natural history of MCD is *correct*?
 - Seventy-five percent of children presenting with MCD never relapse.
 - Most patients with MCD will undergo one or more relapses.
 - End-stage kidney disease is seen in up to 30% patients who present as adults.
 - MCD resolves with puberty.
 - Patients requiring kidney replacement therapy in MCD rarely recover kidney function.
- Which treatment is most appropriate for a 24-year-old woman with relapsing minimal change disease hoping to conceive in the next year?
 - Tacrolimus
 - Rituximab
 - Mycophenolate mofetil
 - Cyclophosphamide
 - Adalimumab
- Which of the following statements regarding complications of MCD is *true*?
 - Hyperlipidemia is a rare complication of MCD.
 - Thrombotic events in MCD are usually arterial.
 - The enhanced risk of infection in MCD is only due to the immunosuppressive therapy seen.
 - AKI occurs in most adults.
 - Rituximab can contribute to hypogammaglobulinemia.

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